

Appendix

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Appendix A. Data source and full project link

1. Data source

Link to the Techmap.io dataset on Amazon Web Services:

<https://aws.amazon.com/marketplace/pp/prodview-p2554p3tczbes>

The dataset was accessed in January 2024.

2. Full project link

The full project can be accessed on GitHub.

GitHub repository with all files: https://github.com/sofiyamusina/ai_and_jobs_ec331

Complete code for the project:

https://github.com/sofiyamusina/ai_and_jobs_ec331/blob/main/dissertation_full_code.py

Code for final part (treatment effect estimation) used in the Tabula submission:

https://github.com/sofiyamusina/ai_and_jobs_ec331/blob/main/dissertation_section5_estimation_code.py

The final part code runs on the following dataset:

https://drive.google.com/file/d/1tRwPVcZBGI-SRN3DNWMAjcsXgM_t2eWL/view?usp=sharing

Appendix B. Matching of vacancies to standardised occupations and assignment of LLM scores

In order to estimate the model, every job ad had to be assigned an LLM exposure score that was, depending on the specification, to represent the “intensity” of the treatment by the “ChatGPT shock”, or to define a cutoff for the treated and control groups.

The only currently available dataset that matches job titles to LLM exposure contains LLM exposure scores for 774 occupations corresponding to 774 US SOC's (Standardised Occupation Codes). Unfortunately, these occupations do not match directly to job titles from the ads in the Techmap.io dataset.

To address this, a matching algorithm was created and implemented. The matching algorithm is inspired by the methodology of researchers from Nesta, who shared their method for assigning online job adverts to occupation groups¹.

Step 1: Vectorization. All the standardised occupation names, as well as all vacancy names, were transformed into 371-dimensional vectors using DistilBert² - an AI model that belongs to the class of pre-trained transformers.

Word or phrase vectorization is a technique that allows to capture semantic meaning in numbers. During the training of a transformer model, a large corpus of text is used to create vectors (also called embeddings) for specific words or expressions. This is based largely on the analysis of the contexts in which a specific word appears.

There are 2 main ways in which a word or phrase can be represented by a vector. The first one is a document-term matrix - a “*mathematical matrix that describes the frequency of terms that occur in a collection of documents*”³. Rows represent documents, columns represent words, and

¹ <https://www.nesta.org.uk/project-updates/soc-assignment-ojo/>

² https://huggingface.co/docs/transformers/en/model_doc/distilbert

³ <https://bookdown.org/Maxine/tidy-text-mining/tidying-a-document-term-matrix.html>

matrix values refer to the number of term appearances of that document. A simple example of a document-term matrix is:

D1. This software engineer writes great code.

D2. Full-stack developers write cleaner code.

D3. This finance manager role requires excel skills.

	<i>this</i>	<i>software</i>	<i>engineer</i>	<i>write(s)</i>	<i>great</i>	<i>code</i>	<i>full-stack</i>	<i>developers</i>	<i>clean</i>	<i>finance</i>	<i>manager</i>	<i>role</i>	<i>requires</i>	<i>excel</i>	<i>skills</i>
<i>D1</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
<i>D2</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>1</i>	<i>0</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
<i>D3</i>	<i>1</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>1</i>	<i>1</i>

The above document-term matrix demonstrates how row vectors encode meaning: expressions “software engineer” and “developers” share a common context with “write” and “code”, words common to the programming profession. A creation of this matrix is one of the steps in transforming expressions into embeddings.

The second way, often derived at later stages of the process, is representing a words with a vector of “score” along different dimensions of meaning, for instance in the case of below row vectors:

	<i>Masculinity</i>	<i>Royalty</i>	<i>Youth</i>	<i>Human nature</i>	<i>Modernity</i>
<i>King</i>	<i>1</i>	<i>1</i>	<i>0.2</i>	<i>1</i>	<i>0.2</i>
<i>Princess</i>	<i>0</i>	<i>0.9</i>	<i>0.7</i>	<i>1</i>	<i>0.2</i>
<i>Crown</i>	<i>0.6</i>	<i>0.8</i>	<i>0.1</i>	<i>0</i>	<i>0.3</i>

Using a combination of these 2 processes, DistilBert can transform both vacancy names and standardised occupation names into vectors. DistilBert is a pre-trained model, such that it is already trained on a large corpus of text and already contains a collection of words matched with embeddings.

Step 2: Matching. Every vacancy name is matched to an SOC by creating a pair of corresponding embeddings that maximises their cosine similarity. In algebra, cosine similarity is

used to compute the distance between vectors. In language processing context, therefore, 2 embeddings that are more semantically alike have a smaller distance and a larger cosine similarity value.

During the matching process in this project, cosine similarity values are calculated for all vacancy-standardised occupation pairs. Each vacancy title is assigned a standardised occupation name match based on the standardised occupation name with which cosine similarity is the highest.

Then, AI exposure scores can be assigned to vacancy names corresponding to SOC, as seen in the below excerpt:

From Techmap.io data	From Felten et al. (2023)	
Vacancy name	SO name	LLM exposure
Sales & Administrative Assistant (m/f)	Sales managers	1.294
IT Analyst Developer - Application Integration	Software developers, applications	0.882
System & DevOps engineer (m/f)	Software developers, system software	1.166
Tax compliance officer: comfortable position	Compliance officers	0.572
Senior funds lawyer: transforming law firm	Lawyers	1.454
Account Payable Specialist	Financial specialists, all other	1.253
Accounting Manager - RE Investment firm	Financial managers	1.295
Luxembourg - Service Architect (F/M)	Architectural and engineering managers	0.647

As seen from above, the matching procedure is not completely accurate. However, the overall accuracy of the matches make it a suitable LLM exposure assignment mechanism.

Appendix C. Treatment group assignment for ATE estimation: LLM exposure cutoffs

Variable	Cutoff score, defined by the place of job LLM exposure in LLM exposure distribution
Experience, extensive margin	50th percentile
Experience, intensive margin	50th percentile
Education, extensive margin	50th percentile
Cognitive skills, extensive margin	mean
Cognitive skills, intensive margin	mean
Creativity, extensive margin	33th percentile
Creativity, intensive margin	33th percentile
Interpersonal skills, extensive margin	50th percentile
Interpersonal skills, intensive margin	50th percentile
Leadership skills, extensive margin	50th percentile
Leadership skills, intensive margin	50th percentile
Time management skills, extensive margin	50th percentile
Time management skills, intensive margin	mean
Motivation, extensive margin	33th percentile
Motivation, intensive margin	33th percentile
Flexibility, extensive margin	50th percentile
Flexibility, intensive margin	50th percentile
Independence, extensive margin	67th percentile
Independence, intensive margin	67th percentile

Appendix D. Creation of outcome variables: skill dictionaries used for keyword search

Skill group	Dictionary content
Interpersonal	["communication", "communicate", "communicating", "interpersonal", "presentation", "presentations", "present", "presenting", "influence", "persuade", "relationships", "collaborate", "collaborative", "collaboration", "team", "teamwork", "part of team", "part of a team", "team spirit", "team player"]
Leadership	["lead", "leadership", "project management", "manage teams", "manage a team", "managing a team", "motivate", "coordinate", "coordination"]
Cognitive	["analysis", "analytical", "analyse", "analyze", "analyzing", "analysing", "problem", "problems", "problem solving", "problem-solving", "problem resolution", "solution", "solution-oriented", "solution oriented", "solutions", "problem solver", "decision-making", "decision making", "decisions", "judgment", "judgements", "diagnose", "suggest", "suggestions", "propose", "summarise", "summarize", "summarizing", "summarising", "interpret", "interpreting", "structurings"]
Time management	["plan", "planning", "organize", "organise", "organization", "organizational", "organisation", "organisational", "organized", "well-organized", "organised", "deadlines", "workload", "priorities", "prioritise", "prioritize", "timekeeping", "time management", "time-management", "simultaneous", "multiple", "timely"]
Creativity	["open-mindedness", "open-minded", "open minded", "creativity", "creative", "create", "creating", "innovative", "innovation", "innovate", "entrepreneurial", "ideas"]
Motivation	["motivated", "interest", "interested", "committed", "ambitious", "initiative", "initiatives", "proactive", "proactively", "proactivity", "pro-active", "passion", "passionate", "hard working", "hard-working", "hardworking"]
Flexibility	["adapt", "ambiguous", "adaptable", "adaptability", "versatile"]
Independence	["independent", "independently", "self-starter", "autonomous", "autonomy", "autonomously", "responsibility", "responsible", "ownership"]

Appendix E. Estimation results excluded from main body

1. Estimation results

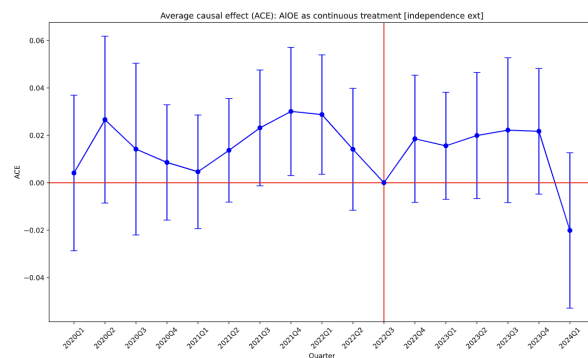
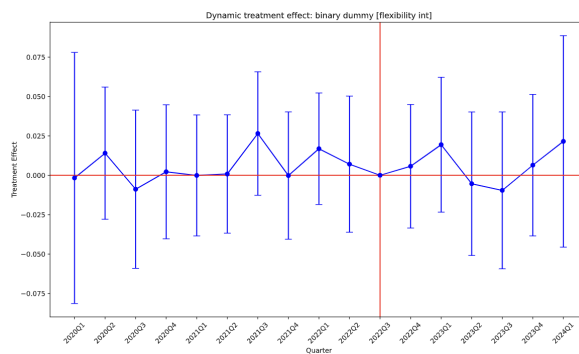
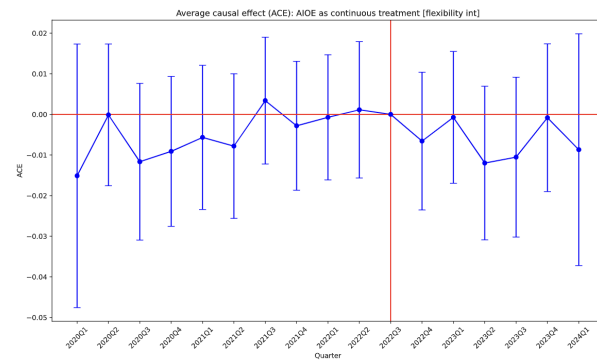
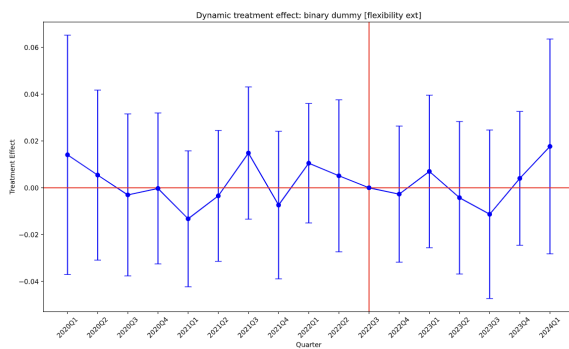
Link to complete estimation results for the **discrete** specification:

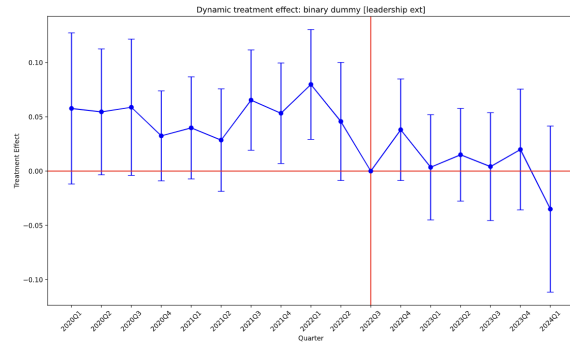
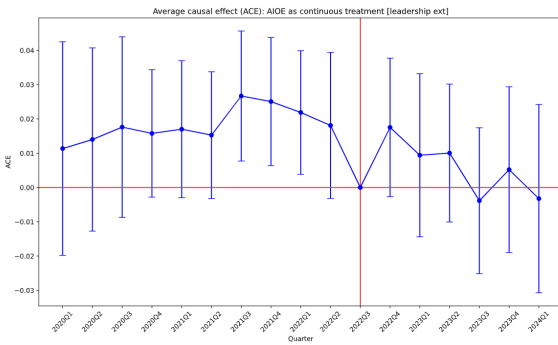
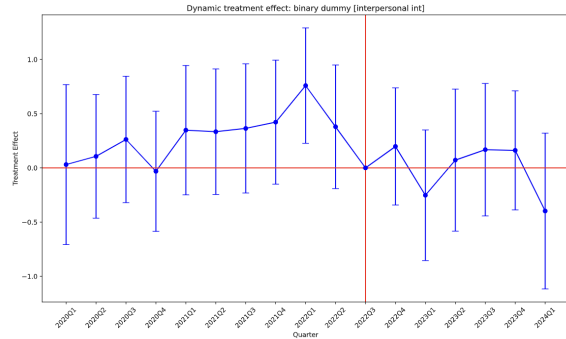
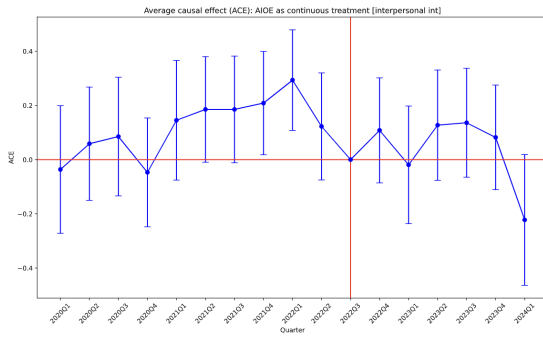
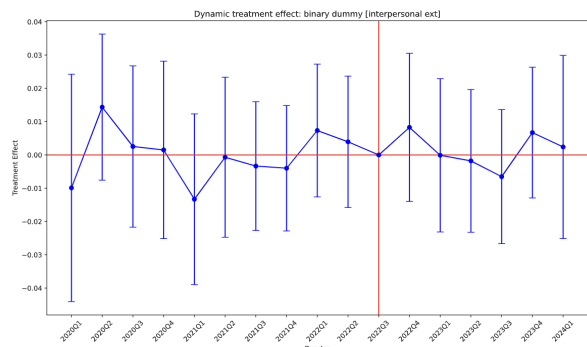
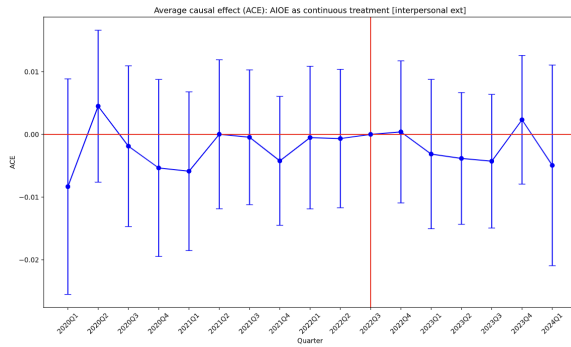
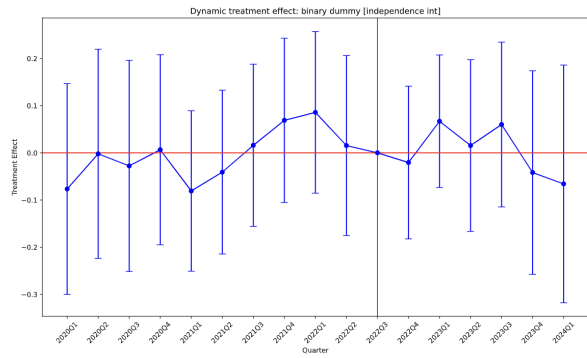
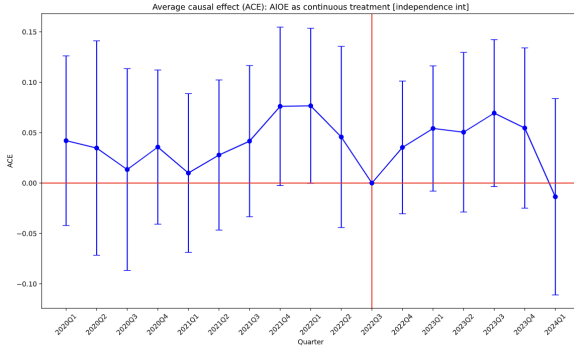
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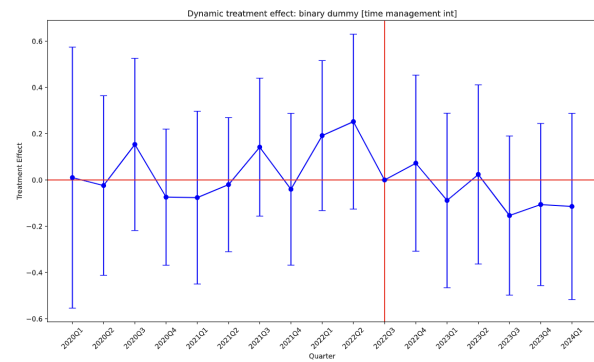
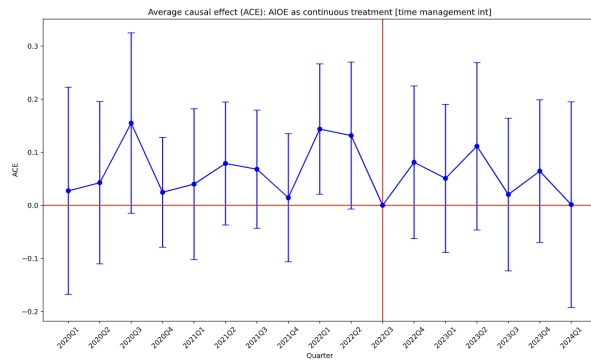
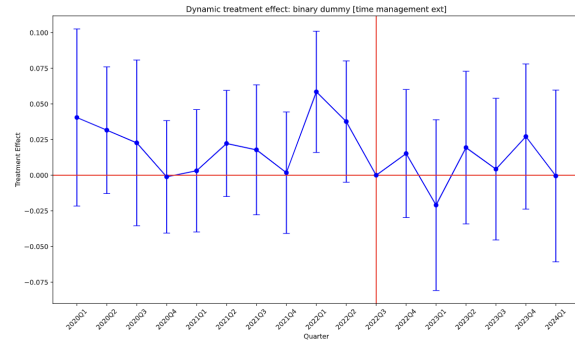
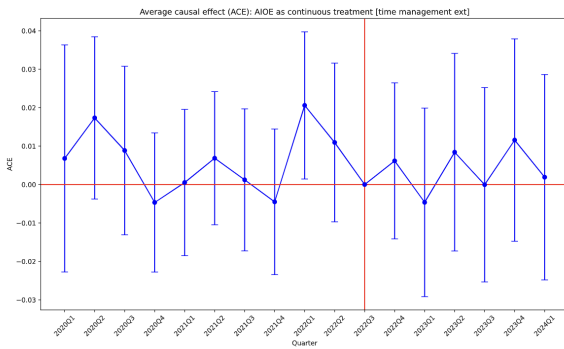
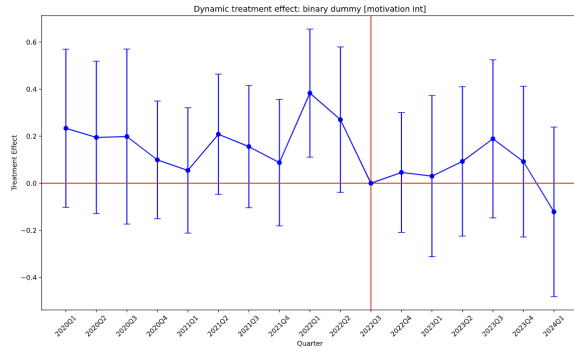
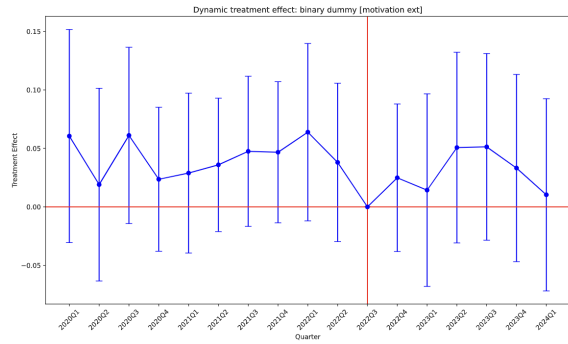
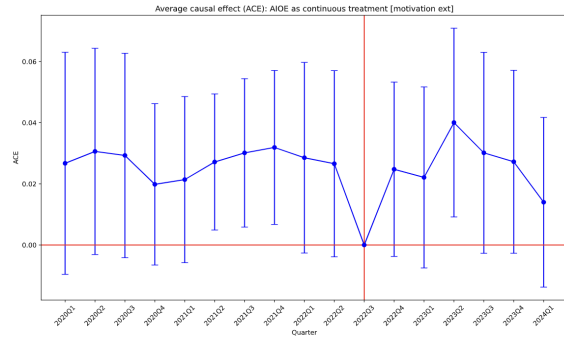
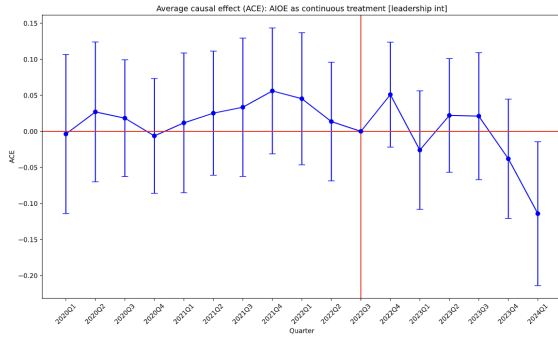
Link to complete estimation results for the **continuous** specification:

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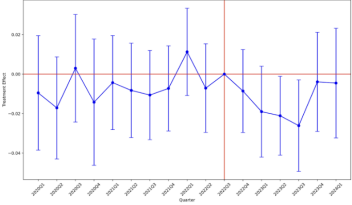
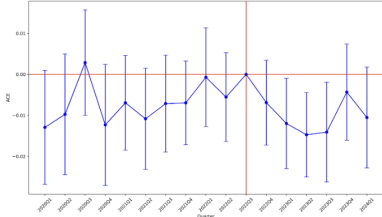
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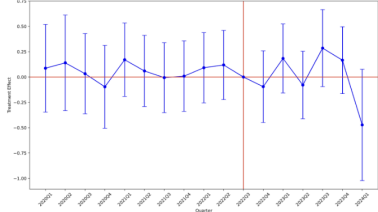
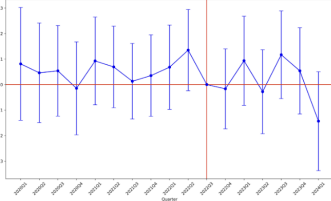


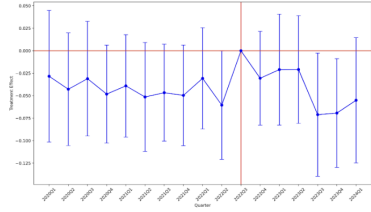
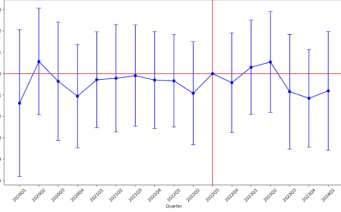




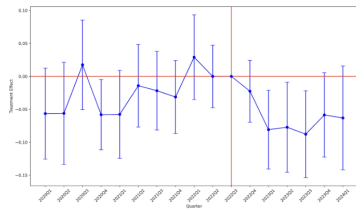
3. Point estimate details for graphs included in main body

ATE estimates for Experience, extensive margin						ACR estimates for Experience, extensive margin					
											
	Coefficient	Std. error	p	Confidence Interval			Coefficient	Std. error	p	Confidence Interval	
β_{12}	-0.009	0.011	0.424	-0.030	0.012	β_{12}	-0.007	0.005	0.192	-0.017	0.003
β_{13}	-0.019	0.012	0.105	-0.042	0.004	β_{13}	-0.012	0.006	0.033	-0.023	-0.00
β_{14}	-0.021	0.010	0.038	-0.041	-0.001	β_{14}	-0.015	0.005	0.005	-0.025	-0.004
β_{15}	-0.026	0.012	0.027	-0.049	-0.003	β_{15}	-0.014	0.006	0.023	-0.026	-0.002
β_{16}	-0.004	0.013	0.759	-0.029	0.021	β_{16}	-0.004	0.006	0.470	-0.016	0.007
β_{17}	-0.005	0.014	0.749	-0.032	0.023	β_{17}	-0.011	0.006	0.093	-0.023	0.002

ATE estimates for Experience, intensive margin						ACR estimates for Experience, intensive margin					
											
	Coefficient	Std. error	p	Confidence Interval			Coefficient	Std. error	p	Confidence Interval	
β_{12}	-0.095	0.181	0.599	-0.449	0.259	β_{12}	-0.017	0.080	0.835	-0.173	0.140
β_{13}	0.183	0.174	0.291	-0.157	0.524	β_{13}	0.094	0.089	0.293	-0.081	0.268
β_{14}	-0.079	0.170	0.644	-0.412	0.255	β_{14}	-0.028	0.084	0.740	-0.192	0.137
β_{15}	0.286	0.194	0.141	-0.094	0.666	β_{15}	0.117	0.088	0.183	-0.055	0.289
β_{16}	0.166	0.168	0.323	-0.163	0.496	β_{16}	0.054	0.086	0.532	-0.115	0.223
β_{17}	-0.473	0.280	0.092	-1.022	0.077	β_{17}	-0.143	0.099	0.147	-0.337	0.050

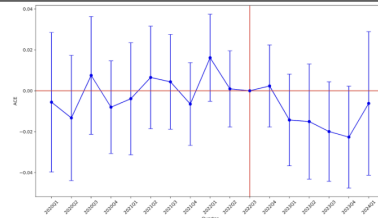
ATE estimates for Education, extensive margin						ACR estimates for Education, extensive margin					
											
	Coefficient	Std. error	p	Confidence Interval			Coefficient	Std. error	p	Confidence Interval	
β_{12}	-0.031	0.027	0.251	-0.083	0.022	β_{12}	-0.004	0.012	0.720	-0.028	0.020
β_{13}	-0.021	0.031	0.505	-0.083	0.041	β_{13}	0.003	0.011	0.791	-0.019	0.025
β_{14}	-0.021	0.031	0.494	-0.081	0.039	β_{14}	0.005	0.012	0.651	-0.018	0.029
β_{15}	-0.071	0.035	0.042	-0.140	-0.003	β_{15}	-0.008	0.014	0.538	-0.035	0.018
β_{16}	-0.069	0.031	0.025	-0.130	-0.009	β_{16}	-0.012	0.012	0.323	-0.034	0.011
β_{17}	-0.055	0.036	0.121	-0.125	0.015	β_{17}	-0.008	0.014	0.569	-0.036	0.020

ATE estimates for Cognitive skills, extensive margin



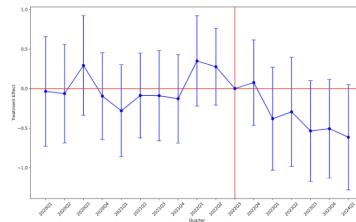
	Coefficient	Std. error	p	Confidence Interval	
β_{12}	-0.023	0.024	0.347	-0.070	0.024
β_{13}	-0.081	0.030	0.008	-0.141	-0.021
β_{14}	-0.077	0.035	0.027	-0.146	-0.009
β_{15}	-0.089	0.034	0.009	-0.154	-0.022
β_{16}	-0.059	0.033	0.073	-0.122	0.005
β_{17}	-0.063	0.040	0.116	-0.142	0.016

ACR estimates for Cognitive skills, extensive margin



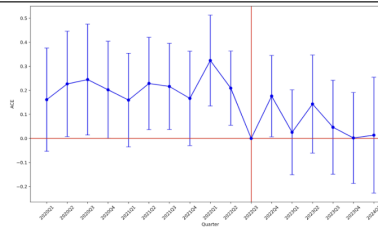
	Coefficient	Std. error	p	Confidence Interval	
β_{12}	0.002	0.010	0.819	-0.018	0.022
β_{13}	-0.014	0.011	0.211	-0.037	0.008
β_{14}	-0.015	0.014	0.294	-0.043	0.013
β_{15}	-0.020	0.012	0.108	-0.044	0.004
β_{16}	-0.023	0.013	0.075	-0.048	0.002
β_{17}	-0.006	0.018	0.728	-0.041	0.029

ATE estimates for Cognitive skills, intensive margin



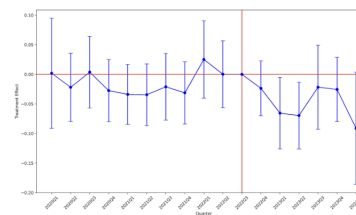
	Coefficient	Std. error	p	Confidence Interval	
β_{12}	0.076	0.275	0.783	-0.463	0.615
β_{13}	-0.380	0.332	0.252	-1.032	0.271
β_{14}	-0.294	0.352	0.404	-0.984	0.396
β_{15}	-0.535	0.324	0.099	-1.171	0.101
β_{16}	-0.507	0.318	0.111	-1.129	0.116
β_{17}	-0.617	0.339	0.070	-1.280	0.050

ACR estimates for Cognitive skills, intensive margin



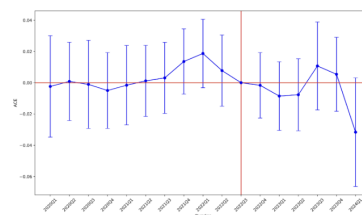
	Coefficient	Std. error	p	Confidence Interval	
β_{12}	0.176	0.086	0.041	0.007	0.345
β_{13}	0.026	0.090	0.777	-0.121	0.202
β_{14}	0.143	0.104	0.169	-0.061	0.347
β_{15}	0.047	0.100	0.640	-0.149	0.241
β_{16}	0.002	0.097	0.984	-0.187	0.191
β_{17}	0.014	0.123	0.912	-0.227	0.255

ATE estimates for Creativity, extensive margin

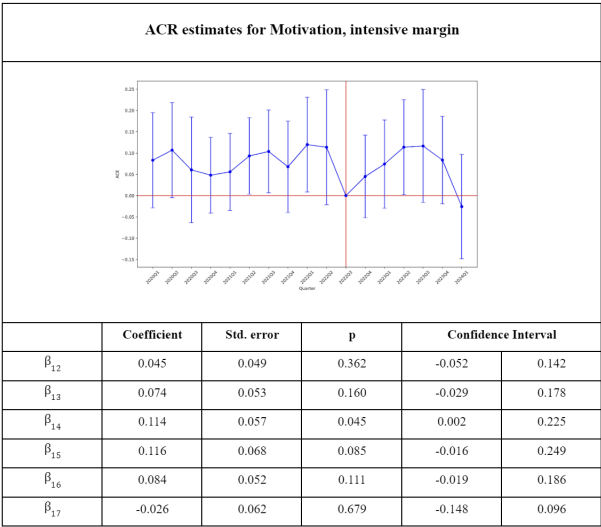
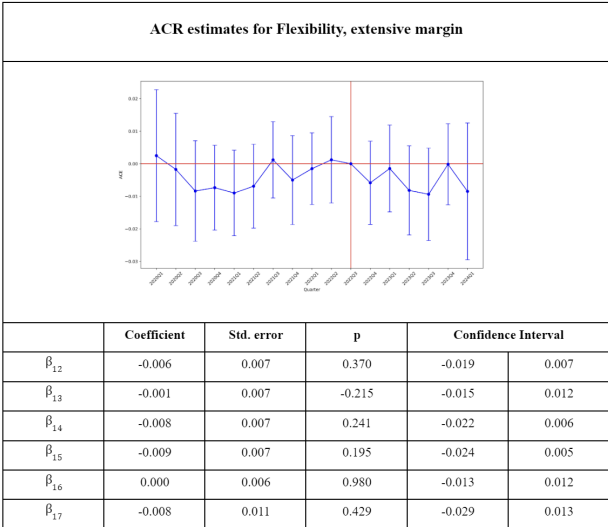
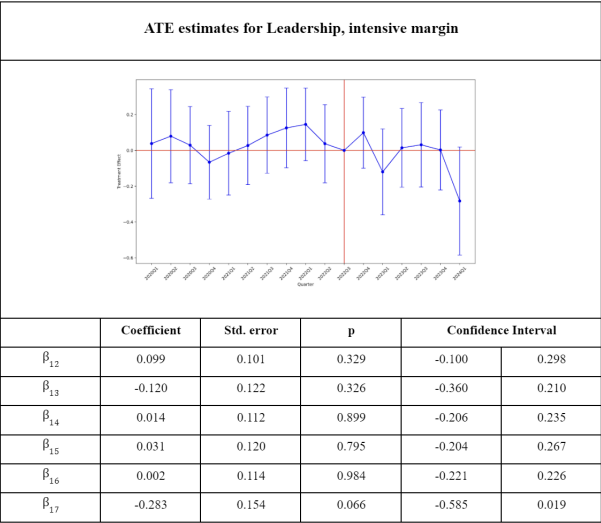
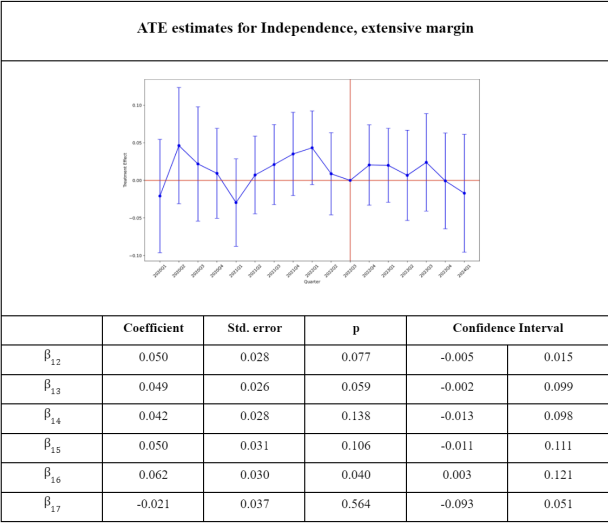
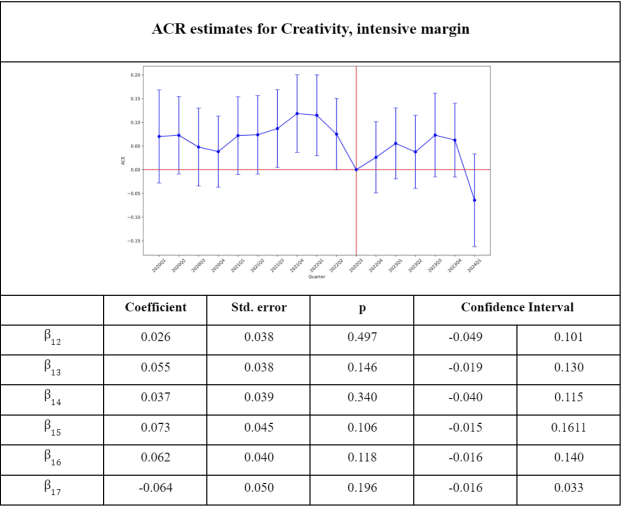
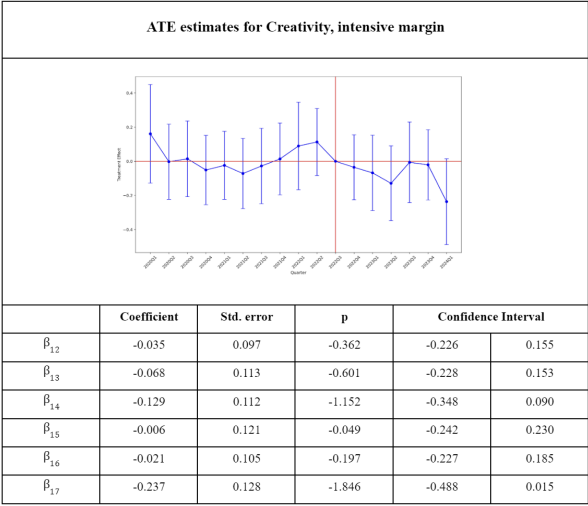


	Coefficient	Std. error	p	Confidence Interval	
β_{12}	-0.024	0.024	0.315	-0.070	0.023
β_{13}	-0.067	0.031	0.033	-0.126	-0.005
β_{14}	-0.070	0.029	0.015	-0.126	-0.013
β_{15}	-0.022	0.036	0.549	-0.093	0.049
β_{16}	-0.027	0.028	0.356	-0.080	0.029
β_{17}	-0.091	0.048	0.059	-0.186	0.004

ACR estimates for Creativity, extensive margin



	Coefficient	Std. error	p	Confidence Interval	
β_{12}	-0.002	0.011	0.878	-0.023	0.019
β_{13}	-0.009	0.011	0.446	-0.030	0.013
β_{14}	-0.008	0.012	0.513	-0.031	0.015
β_{15}	0.011	0.014	0.454	-0.017	0.039
β_{16}	0.005	0.012	0.654	-0.018	0.029
β_{17}	-0.032	0.018	0.074	-0.066	0.003



Appendix F. Comparison of R^2 for specifications with and without control variables

Specification	Discrete Treatment		Continuous Treatment	
Variable	R2, no controls	R2, with source control	R2, no controls	R2, with source control
Experience, extensive margin	0.002	0.005	0.049	0.051
Experience, intensive margin	0.003	0.004	0.077	0.078
Education, extensive margin	0.006	0.010	0.058	0.061
Cognitive skills, extensive margin	0.005	0.017	0.090	0.097
Cognitive skills, intensive margin	0.006	0.023	0.131	0.140
Creativity, extensive margin	0.008	0.029	0.084	0.100
Creativity, intensive margin	0.006	0.025	0.105	0.118
Interpersonal skills, extensive margin	0.004	0.006	0.041	0.039
Interpersonal skills, intensive margin	0.010	0.021	0.076	0.084
Leadership skills, extensive margin	0.004	0.013	0.047	0.052
Leadership skills, intensive margin	0.006	0.014	0.084	0.089
Time management skills, extensive margin	0.003	0.008	0.048	0.052
Time management skills, intensive margin	0.003	0.011	0.081	0.087
Motivation, extensive margin	0.004	0.015	0.031	0.041
Motivation, intensive margin	0.006	0.022	0.055	0.068
Flexibility, extensive margin	0.003	0.004	0.021	0.022
Flexibility, intensive margin	0.004	0.005	0.020	0.021
Independence, extensive margin	0.004	0.007	0.032	0.034
Independence, intensive margin	0.004	0.007	0.031	0.034